



Alberta Public Health Disease Management Guidelines

Listeriosis



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Alberta Health

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Case Definition

Confirmed Case

Clinical evidence of invasive disease^(A) with laboratory confirmation of infection:

- Isolation of *Listeria monocytogenes* from a normally sterile site (e.g., blood or cerebrospinal fluid [CSF] or, less commonly, joint, pleural, or pericardial fluid)

OR

- In the setting of miscarriage or stillbirth, isolation of *L. monocytogenes* from placental or fetal tissue (including amniotic fluid and meconium).

Probable Case

Clinical evidence of invasive disease^(A) in a person who is epidemiologically linked to a laboratory-confirmed case or to a confirmed source.^(B)

^(A) Clinical evidence of invasive disease usually manifests as meningitis or bacteremia. Infection during pregnancy may result in fetal miscarriage or stillbirth, or neonatal meningitis or bacteremia. Other manifestations can also be observed, especially in immunosuppressed individuals and the elderly.

^(B) This probable case definition is provided as a guideline to assist with case finding and public health management, and should not be reported to Alberta Health.

Reporting Requirements

Physicians/Health Practitioners and Others

A physician, health practitioner or others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed cases in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

- All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:
 - Chief Medical Officer of Health (CMOH) (or designate), and
 - MOH (or designate) of the zone.
- All isolates shall be sent to the Public Health Laboratories (ProvLab) for further characterization.

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days) including:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Listeria monocytogenes are small gram-positive, non-spore forming, aerobic bacilli. Human infections are usually caused by serogroups 1/2a, 1/2b, and 4b. The bacteria survive well in soil, water, food, and feces. They are able to grow at low temperatures (3–45°C) and are resistant to freezing and drying.

Clinical Presentation

Listeriosis is a bacterial infection. The clinical manifestations of infection range from mild (gastroenteritis) to severe. Typical symptoms include fever, muscle aches, and on occasion, nausea and vomiting. The bacteria may infect the brain and the membrane lining the brain causing meningoen­cephalitis. The onset of meningoen­cephalitis may be sudden with fever, intense headache, nausea, and vomiting. Complications include endocarditis, and internal and external abscesses. Direct contact with infectious material or soil contaminated with infected animal feces can result in papular lesions on hands and arms. Asymptomatic fecal carriage occurs in approximately 10% of cases.

Infected pregnant women may have minimal symptoms typically characterized by a mild flu-like illness and may unknowingly pass the illness to the unborn child. Infection during pregnancy may lead to premature delivery, infection of the newborn or stillbirth. The infant may develop meningitis. Thirty per cent of newborn infections are fatal. The case-fatality rate is 50% if the onset of illness occurs within the first four days of life.

Diagnosis

The diagnosis is confirmed by isolation of the bacteria from CSF, blood, amniotic fluid, placenta, meconium, lochia, gastric washings, and other sites of infection. *Listeria monocytogenes* can be readily isolated from normally sterile sites on routine media. Care needs to be taken to ensure the culture is distinguished from other gram-positive rods, such as diphtheroids.

Treatment

- Immediate treatment is essential.
- Penicillin and ampicillin are the antibiotics of choice, given alone or together with aminoglycosides.
- TMP-SMX or erythromycin may be used in cases of penicillin allergy.
- Cephalosporins, including third-generation, are not effective in the treatment of clinical listeriosis.
- Tetracycline resistance of the organism has been observed.

Reservoir

Listeria monocytogenes is ubiquitous. It has been isolated in soil, dust, water, and foods. The primary reservoirs are soil and decomposing organic matter. Other reservoirs include the intestinal tract of domestic and wild mammals, fowl, and, on occasion, humans. Infection in foxes may produce an encephalitis simulating rabies. Asymptomatic fecal carriage occurs in man (10%) and animals. The bacteria are frequently found in free-living water and mud. Seasonal use of silage as feed is frequently followed by an increased incidence of listeriosis in animals.

Transmission

Humans commonly contract listeriosis by eating food contaminated with the bacterium *L. monocytogenes*. Livestock become infected after ingesting contaminated silage or water. Vegetables and fruit may become contaminated from the soil or from manure used as a fertilizer. The bacterium has been found in raw foods, such as uncooked meats and vegetables, as well as in processed foods that become contaminated after processing, such as soft cheeses and cold cuts. Unpasteurized milk or

food made from unpasteurized milk may contain the bacterium. Taking antacids may promote infection. In neonates, transmission may occur from mother to fetus in utero or during passage through the infected birth canal. Sexual contact and inhalation of the organism is possible. Nursery outbreaks have also been reported.^(1,2)

Incubation Period

The incubation period ranges from three to 70 days, with the median incubation period estimated to be three weeks.

Period of Communicability

Infected persons may retain the bacteria in their stools for several months. Mothers of infected babies may continue to shed the bacteria in vaginal discharges and urine for seven to ten days after delivery.

Host Susceptibility

The disease generally affects the elderly, pregnant women, and individuals who are immunosuppressed. Immunocompromised persons are most likely to become infected with *L. monocytogenes*. Pregnant women are about 17 times more likely than other healthy adults to get listeriosis. Persons with AIDS are almost 300 times more likely to get listeriosis than persons with normal immune function. In addition, at increased risk are:

- newborns,
- persons with cancer, diabetes or kidney disease,
- individuals taking corticosteroids and/or antacids, and
- organ transplant recipients.^(1,2)

Incidence

General

Listeriosis occurs worldwide. Illness is rare and most infections are asymptomatic. In the USA, the rate of listeriosis has been reported as 1/200,000. Typically infection occurs sporadically; however, outbreaks can occur in all seasons. It is a serious disease when it affects pregnant women, newborns, and adults with weakened immune systems. Pregnant women account for about 30% of all cases. Listeriosis causes meningoencephalitis and/or septicemia in newborns and adults, and abortion in pregnant women. In non-pregnant adults, infection occurs mainly after age 40. Asymptomatic infections probably occur at all ages.

Canada

In Canada, listeria has been reportable since 1990. From 1990 to 1999, the rate remained below 1/100,000, ranging from 0.18/100,000 in 1992 to 0.34/100,000 in 1998. Infants less than one year of age are the most commonly reported age group – in 1998, a rate of 1.55/100,000 was reported. This group is followed closely by persons over the age of 60 years (rate of 1.47/100,000).⁽²⁾

Alberta

In Alberta the rate has remained below 1/100,000 population for the period 1990 to 2004. In 1999, the most common age group affected was individuals over the age of 60 years with a rate of 2.02/100,000, followed by those 40 to 59 years with a rate of 0.39/100,000.^(2,3)

From 1985 to 2003 there were 121 cases of listeriosis reported in Alberta. The most common source of infection reported on Notifiable Disease Reports from health regions includes unknown (101), food (16), vertical transmission (one), animal source (one), travelling abroad (one), and home (one).

Public Health Management

Key Investigation

- Determine the possible source of the infection taking into consideration the incubation period, reservoir, and mode of transmission. Assessment may include:
 - obtaining a detailed food history, focusing on foods potentially contaminated with *L. monocytogenes* such as unpasteurized cheese, milk, yogurt, deli meats, raw or undercooked wieners, fresh unwashed garden vegetables, and pâté.
 - collecting food samples for culture if available.⁽⁴⁾
 - Contact the microbiologist on-call for advice on how to collect and transport food samples,
 - Suspect food samples are referred for culture to the Listeriosis Reference Laboratory in Ottawa.
 - Check the [Canada Food Inspection Agency website](#) for recalled foods.
 - determining a history of contact with infective materials such as aborted fetuses on farms, sick or dead animals (especially sheep with encephalitis), animal feeds, animal compost, and manure.
- Determine history of daycare or hospital exposure.
- Identify potentially contaminated water source.
- Identify others who may have been exposed to the same source.
- Suspected contaminated food may be held to prevent consumption.
- Suspected contaminated food may be destroyed.

Management of a Case

- Supportive treatment.
- Routine practices for hospitalized individuals.
- Prophylactic antibiotics should be administered to asymptomatic newborns if they have short gram-positive rods in meconium.

Management of Contacts

- No public health intervention is required as person-to-person transmission rarely occurs.
- Symptomatic and asymptomatic contacts should be investigated if a common source is suspected.

Preventative Measures

- Thoroughly cook raw food from animal sources (e.g., beef, pork, and poultry).
- Thoroughly wash raw vegetables and fruit before eating.
- Keep uncooked meats separate from vegetables, cooked foods, and ready-to-eat foods.
- Avoid consumption of unpasteurized milk or foods made from raw milk.
- Wash hands, knives, and cutting boards after handling uncooked foods.
- Additional recommendations for persons at high risk (those immunocompromised by illness or medications, pregnant women, and elderly persons) include:
 - avoiding **unpasteurized** cheeses (there is no need to avoid pasteurized cheeses, cream cheese, cottage cheese or yogurt),
 - thoroughly reheating (i.e., should be steaming hot) leftovers of ready-to-eat foods (e.g., hot dogs) before eating, and
 - avoiding foods or thoroughly reheating cold cuts from delicatessen counters - especially pregnant women and immunocompromised persons, although the risk is relatively low.

- Education of veterinarians and farmers to take proper precautions in handling aborted fetuses, and sick or dead animals.
- Identification of clusters of a possible common source.
- Avoid the use of untreated manure on vegetable crops.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
September 2021	General	• Updated Template • Diagnosis and Treatment section moved to Epidemiology • Updated web links

References

- (1) Government of Canada. *Pathogen Safety Data Sheets: Infectious Substances – Listeria monocytogenes*. April 2012.
<https://www.canada.ca/en/public-health/services/laboratory-biosafety-biosecurity/pathogen-safety-data-sheets-risk-assessment/listeria-monocytogenes.html>
- (2) Public Health Agency of Canada. Notifiable Diseases On-line — Listeriosis. July 2003.
http://dsol-smed.hc-sc.gc.ca/dsol-smed/ndis/index_e.html
- (3) Alberta Health and Wellness, Disease Control and Prevention. *Communicable Disease Reporting System*. August 2002.
- (4) *Guide to Services*. Provincial Laboratory for Public Health (Microbiology) and Capital Health Medical Microbiology Laboratory. December 2004.